1 Quick refresher on spin

Spin can be described with the operators $\hat{S}_x, \hat{S}_y, \hat{S}_z$ that obey the following commutation relations:

$$\boxed{[\hat{S}_j, \hat{S}_k] = i\hbar\,\epsilon_{jk\ell}\hat{S}_\ell} \qquad (\hbar = 1), \tag{1}$$

with $i, j, k = x, y, z = 1, 2, 3$ and using the Einstein summation convention. One can define raising and lowering operators as

$$\boxed{\hat{S}_\pm = \hat{S}_x \pm i\hat{S}_y} \qquad \text{with} \qquad \boxed{[S_-, S_+] = -2S_z}. \tag{2}$$

Proof.

$$\begin{aligned}
[S_-, S_+] &= [S_x - iS_y,\ S_x + iS_y] \\
&= [S_x, S_x] + i[S_x, S_y] - i[S_y, S_x] + [S_y, S_y] \\
&= 2i[S_x, S_y] \overset{(1)}{=} 2i \cdot i\,\epsilon_{12\ell}S_\ell = -2\,\epsilon_{12\ell}S_\ell = -2S_z,
\end{aligned} \tag{3}$$

where the first step used the bilinearity of the commutator and the last step uses $\epsilon_{jk\ell} = 0$ if two indices are equal, leaving $\epsilon_{123}S_3 = S_z$. □

For spin-$\frac{1}{2}$ particles, one can write

$$S_j = \frac{\hbar}{2}\,\sigma_j \qquad \text{with } \sigma_j \text{ the Pauli matrices, } j = x, y, z, \tag{4}$$

and often again $\hbar = 1$.